

CSP-571 Project Proposal and Outline

Project Title: Healthcare Data Analysis: Predicting Diabetes Risk

Course: CSP 571 - Data Preparation & Analysis

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1. Project Proposal

1.1. Project Description and Research Goal

Diabetes is a widespread chronic disease that requires early detection for better management. This project aims to develop a simple yet effective machine learning model to predict diabetes risk based on medical diagnostic measurements. We will use Logistic Regression, a widely used classification algorithm, to assess diabetes risk in individuals.

1.2. Research Questions

1. What are the key medical indicators that contribute to diabetes risk?
2. How well does Logistic Regression perform in predicting diabetes compared to a baseline model?
3. What preprocessing steps improve model accuracy?

1.3. Methodology

Data Collection & Preprocessing

- Dataset: PIMA Indians Diabetes Dataset from the UCI Machine Learning Repository.
- Handling Missing Values: Mean imputation for missing values.
- Feature Scaling: Normalize numerical features where needed.
- Train-Test Split: 80% training and 20% testing.

Exploratory Data Analysis (EDA)

- Visualize distributions using histograms and boxplots.
- Identify correlations among features using Pearson correlation.
- Detect and handle outliers using Z-score analysis.

Model Development

- Implement Logistic Regression as the primary model.
- Use hyperparameter tuning to optimize model performance.
- Evaluate using Accuracy, Precision, Recall, and F1-score.

1.4. Metrics for Evaluating Analysis Results

- Accuracy: Measures overall correctness.
- Precision & Recall: Evaluate the balance between false positives and false negatives.
- F1-score: Combines precision and recall for a balanced metric.

2. Project Outline

2.1. Literature Review and Related Work

- Research papers on diabetes prediction using machine learning.
- Studies on the effectiveness of Logistic Regression in medical applications.
- Comparative analyses of different feature selection techniques.

2.2. Data Sources & Description

- Dataset: PIMA Indians Diabetes Dataset.
- Source: UCI Machine Learning Repository.
- Dataset Size: 768 observations, 8 predictor variables, 1 binary outcome variable.
- Feature Descriptions: Pregnancies, Glucose levels, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, Age, Outcome (Diabetes: 1, No Diabetes: 0).

2.3. Data Processing & Pipeline

1. Data Cleaning & Handling Missing Values:

- Use mean imputation for missing values.
- Normalize numerical variables where necessary.

2. Feature Engineering:

- Select features based on correlation analysis.

3. Outlier Detection:

- Use boxplots and statistical methods to detect anomalies.

2.4. Model Selection

- Implement Logistic Regression as the classification model.
- Compare performance with a baseline predictor.
- Optimize model using cross-validation.

2.5. Software, Libraries & Tools

- Programming Language: R
- Libraries Used:
 - tidyverse – Data manipulation and visualization.
 - caret – Machine learning framework for model training and tuning.
 - ggplot2 – Data visualization.

3. Conclusion & Future Scope

This project demonstrates how a simple yet effective Logistic Regression model can be used to predict diabetes risk based on medical diagnostic data. By identifying key risk factors and evaluating model performance, this project contributes to data-driven healthcare solutions.

Future Improvements:

- Expand dataset with additional patient records.
- Incorporate feature engineering techniques for better performance.
- Test additional models for comparison.